

		3	no change (<i>accept stay/remain dark</i>)				B1	[3]
						Total		[11]
7	(a)	(i)	$P = VI$ current = $60/240 = 0.25 \text{ A}$				C1	
		(ii)	$R (= V/I) = 240/0.25$ $= 960 \Omega$				A1	
							M1	
							A0	[3]
	(b)		$R = \rho L/A$ (<i>wrong formula, 0/3</i>)				C1	
			$960 = (7.9 \times 10^{-7} \times L)/(\pi \times \{6.0 \times 10^{-6}\}^2)$				C1	
			$L = 0.137 \text{ m}$				A1	[3]
			<i>(use of $A = 2\pi r$, then allow 1/3 marks only for resistivity formula)</i>					
	(c)		e.g. the filament must be coiled/it is long for a lamp				B1	[1]
			<i>(allow any sensible comment based on candidate's answer for L)</i>					
						Total		[7]
8	(a)		$V/E = R/R_{\text{tot}}$	or	$0.5 = I \times 3900$		C1	
			$1.0/1.5 = R/(R + 3900)$	or	$1.0 = 0.5R/3900$		M1	